

Solution Requirements

Date	07 july 2026
Team ID	
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users log in securely using email and password or institutional credentials.	High
2	Authorization Levels	Two roles: Student (can record/upload audio, view feedback) and Admin (can manage reference answers, view analytics).	High

3	External Interfaces	Integrates with Speech-to-Text API (e.g., Whisper, Azure Speech) and NLP models (e.g., BERT, Sentence Transformers).	High
4	Transaction Processing	Handles audio uploads, text conversion, similarity scoring, and feedback generation in real time.	High
5	Reporting	Generates detailed feedback reports showing similarity scores, missing concepts, and improvement suggestions.	High
6	Business Rules	Each concept explanation is compared against a predefined reference answer; feedback must include at least one improvement suggestion.	Medium
7	Compliance to Laws or Regulations	Adheres to data privacy standards (GDPR-like compliance); stores no personal audio beyond user consent.	High
8	Other	Supports multiple languages for concept explanations and feedback.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Speech-to-text conversion and similarity scoring should complete quickly.	Response time < 3 seconds per audio file
2	Scalability	System should handle multiple concurrent users and large datasets.	Supports up to 10,000 users simultaneously
3	Security & Data Privacy	All data (audio, text, reports) encrypted during storage and transmission.	End-to-end AES-256 encryption
4	Reliability & Availability	System should remain operational with minimal downtime.	99.9% uptime per month
5	Usability & Accessibility	Interface should be intuitive and accessible to all users.	Complies with WCAG 2.1 AA standards
6	Other	Maintainability and portability across cloud platforms.	Modular architecture deployable on Azure, AWS, or GCP